WISH: A Gold-Standard 12-Lead ECG & Motion Wearable Platform

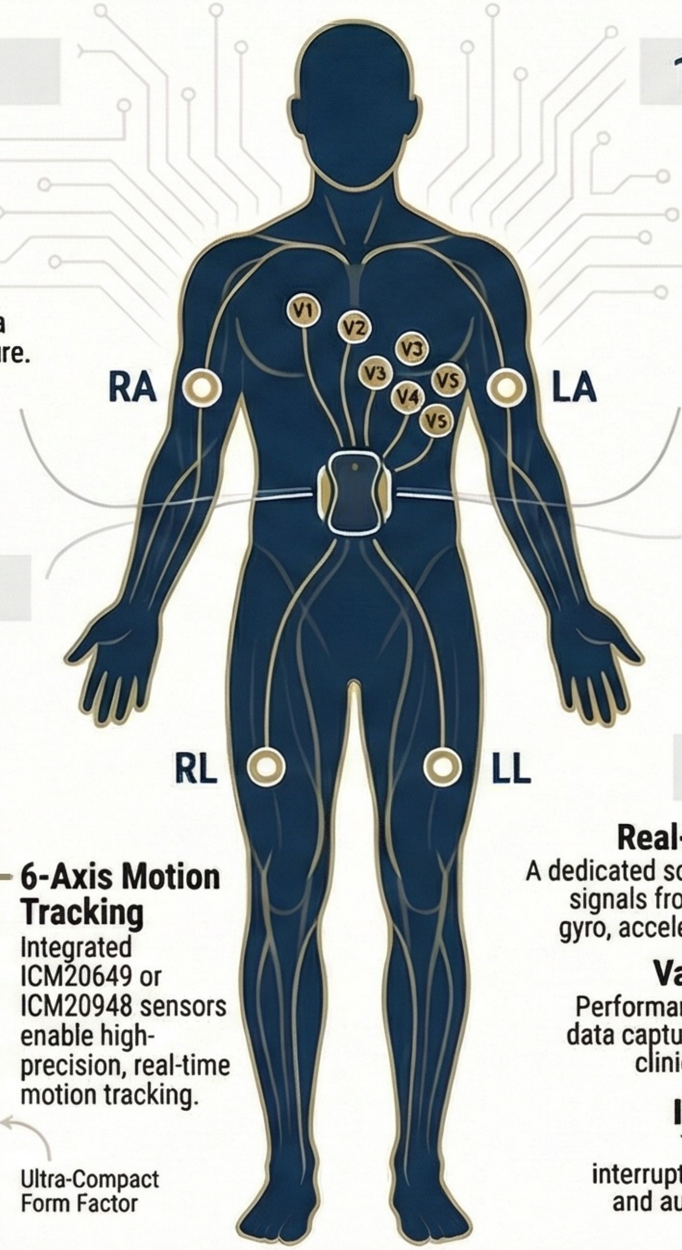
WISH

SYSTEM OVERVIEW & KEY FEATURES

Multi-Modal Sensing Platform
A compact wearable designed for simultaneous capture of biopotential (ECG) and 6-axis motion data (IMU).

12-Lead Gold Standard Monitoring
Supports up to 12 electrode inputs for continuous monitoring with a specialized Right Leg Drive (RLD) feature for high-quality ECG capture.

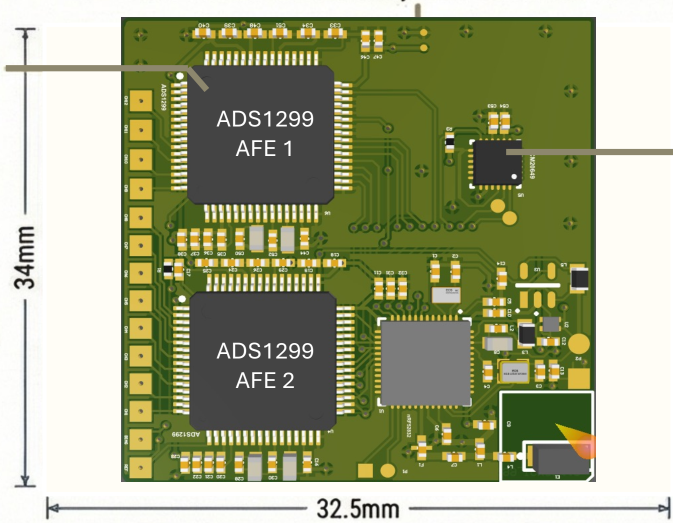
High-Speed BLE Connectivity
The system features Bluetooth Low Energy (BLE) support with up to 2 Mbps bandwidth for real-time data transmission.



HARDWARE ARCHITECTURE

nRF52832 Microcontroller
ARM Cortex M4-based controller that manages the BLE stack and overall system coordination.

ADS1299 AFE 1
Two biopotential Analog Front Ends provide multi-channel simultaneous sampling with ultra-low noise noise signal capture and programmable gain.



6-Axis Motion Tracking
Integrated ICM20649 or ICM20948 sensors enable high-precision, real-time motion tracking.

Ultra-Compact Form Factor

10-ELECTRODE / 12-LEAD CONNECTION PLAN

Electrode Mapping

Electrode	Assignment	Function
V1 – V6	CH1 – CH6 (ADS1299_1)	Precordial Chest Leade
RA / LL	CH7 – CH8 (ADS1299_1)	Right Arm / Left Leg
LA / REF	CH1 – REF (ADS1299_2)	Left Arm / Common Reference
RL (BIAS)	BIAS	Right Leg / Single Bias Drive

Bipolar Lead Derivation
Derives Lead I (LA–RA), Lead II (RA–LL), and Lead III (LL–LA) using standard limb electrode placements.

Unipolar & Augmented Leads
Calculates Wilson Central Terminal (WCT) to derive chest leads (V1-V6) and augmented limb leads (aVR, aVL, aVF).

Clock Synchronization
The circuit utilizes clock sharing between ADS1299 chips to ensure perfectly synchronized measurements across all channels.

REAL-TIME VISUALIZATION & RESULTS

Real-Time Python Dashboard
A dedicated software template visualizes ECG signals from all channels alongside 6-axis gyro, accelerometer, and temperature data.

Validated Against Bioradio
Performance results show the wearable's data capture is comparable to established clinical-grade "Bioradio" equipment.

Intelligent Fault Handling
The application design includes interrupts for data ready signals (DRDY) and automatic wearable shutdown for multiple peripheral failures.